

MAYANK BANSAL

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

TECHNICAL SKILLS

Machine Learning and AI: Scikit-learn, CNNs, Neural Networks, Model Evaluation, Feature Engineering, Exploratory Data Analysis

Deep Learning Frameworks: TensorFlow, PyTorch

Data and Visualization: NumPy, Pandas, Matplotlib, Seaborn

NLP and LLM Systems: TF-IDF, Sentence Transformers, Retrieval Augmented Generation, FAISS, Embeddings

Backend and Deployment: Flask, REST APIs, Gunicorn, Render

Programming Languages: Python, JavaScript, Java, C++, SQL, HTML, CSS

PROJECTS

AI Complaint Triage & Routing System | [GitHub](#)

Jan 2026 – Feb 2026

- **Objective:** Automate complaint handling at scale by building an end-to-end ML triage system routing across 8 departments with urgency prediction and confidence aware outputs
- **Approach:** Designed a hybrid credibility scoring pipeline combining rule-based logic and ML confidence to assess actionability and trigger follow ups
- **System Design:** Engineered a modular Flask REST API with separated inference, credibility logic, and explainable decision flow. Designed and published an 800-sample dataset for multi-class complaint routing and urgency classification on Kaggle.
- **Outcome:** Enabled auditable, feedback driven complaint handling with a foundation for continuous model improvement by retraining

Tech: Python, Flask, **Scikit-learn**, **TF-IDF**, SQLite

PDF Question Answering System (RAG) | [GitHub](#)

Dec 2025 – Jan 2026

- **Objective:** Build a reliable question answering system over unstructured PDFs by grounding LLM responses in retrieved document context
- **Approach:** Implemented a RAG pipeline using document chunking with overlap, FAISS vector similarity search, and Sentence Transformer embeddings
- **System Design:** Engineered prompt level guardrails to reduce hallucinations and enforce explicit 'I don't know' responses for out-of-scope queries
- **Outcome:** Delivered transparent, verifiable answers with source attribution via a Flask backend and HTML, CSS, JavaScript frontend

Tech: **FAISS**, **Sentence Transformers**, Flask, **Gemini API**

WasteSnap – AI-Powered Waste Classification App | [GitHub](#)

May 2025 – May 2025

- **Objective:** Enable automated waste segregation from images to support accurate, sustainable disposal decisions in real world settings
- **Approach:** Trained a MobileNetV2 based multi class image classifier using transfer learning, data augmentation, and fine tuning to handle real world lighting and background variation
- **System Design:** Optimized the model for real time inference while maintaining robustness on high intra class similarity categories
- **Outcome:** Delivered real time recyclability classification with eco-friendly disposal recommendations to promote sustainable waste management

Tech: **MobileNetV2**, Flask, JavaScript, Render

Mood2Mail – Real-Time Email Tone Analyzer | [GitHub](#)

Apr 2025 – Apr 2025

- **Objective:** Help users improve professional communication by detecting and correcting unintended email tone before sending
- **Approach:** Built an NLP based tone classifier using TF-IDF vectorization and **Multinomial Naive Bayes** across five tone categories
- **System Design:** Created and manually annotated a 475-sample real world email dataset and validated performance through preprocessing and train test evaluation, also published the same on Kaggle.
- **Outcome:** Delivered real time tone feedback through a responsive Flask and JavaScript web interface, enabling proactive tone adjustment

Tech: **TF-IDF**, **Naive Bayes**, Flask, JavaScript

ACHIEVEMENTS

- Published a research paper, *Mood2Mail: A Lightweight Real Time Email Tone Analyzer Using Classical Machine Learning*, on Zenodo with DOI, covering dataset creation, TF IDF feature engineering, and model evaluation
- Solved 580+ competitive programming problems on GeeksForGeeks with a 101-day coding streak and a rank of 159 among 39,000+ participants along with 100+ problems on LeetCode

CERTIFICATIONS

- Complete Data Science and Machine Learning, GeeksForGeeks Jun 2025 – Nov 2025
- AI Tools Skill Up, GeeksForGeeks Jun 2025 – Aug 2025

EDUCATION

Lovely Professional University – Bachelor of Technology in Computer Science Engineering

Phagwara, Punjab, India | Aug 2024 – May 2028 | **CGPA: 9.33/10**

St. Joseph's International School – XII Senior Secondary (Physics, Chemistry, Mathematics)

Hisar, Haryana, India | Apr 2023 – Apr 2024